



the full report

A Data Channel over the Find My Network

Hundreds of millions of finder devices relay beacons for anyone. What kind of channel is it?

>99.5 %

deliverability at full power

0

payload bytes ever corrupted

158_s

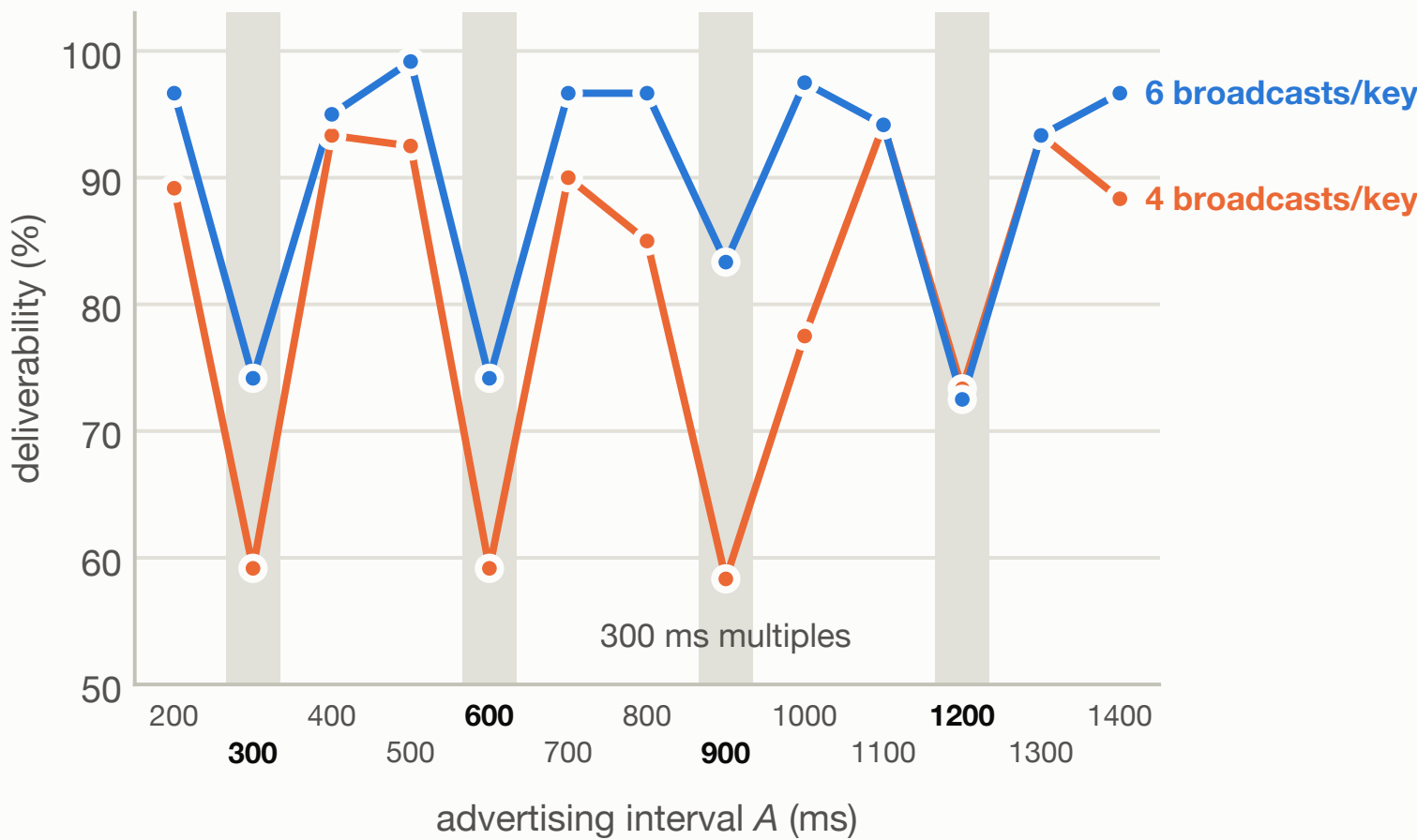
broadcast-to-report delay

0.06–1.2 byte/s

goodput, trading against reliability

Deliverability collapses at every multiple of 300 ms

- Sharp dips at 300, 600, 900, 1200 ms
- **69.3%** there against **92.7%** elsewhere
- Beyond chance: $p < 5 \times 10^{-5}$
- Same dips at 4 and 6 broadcasts per key
- So it tracks the interval, not key rotation

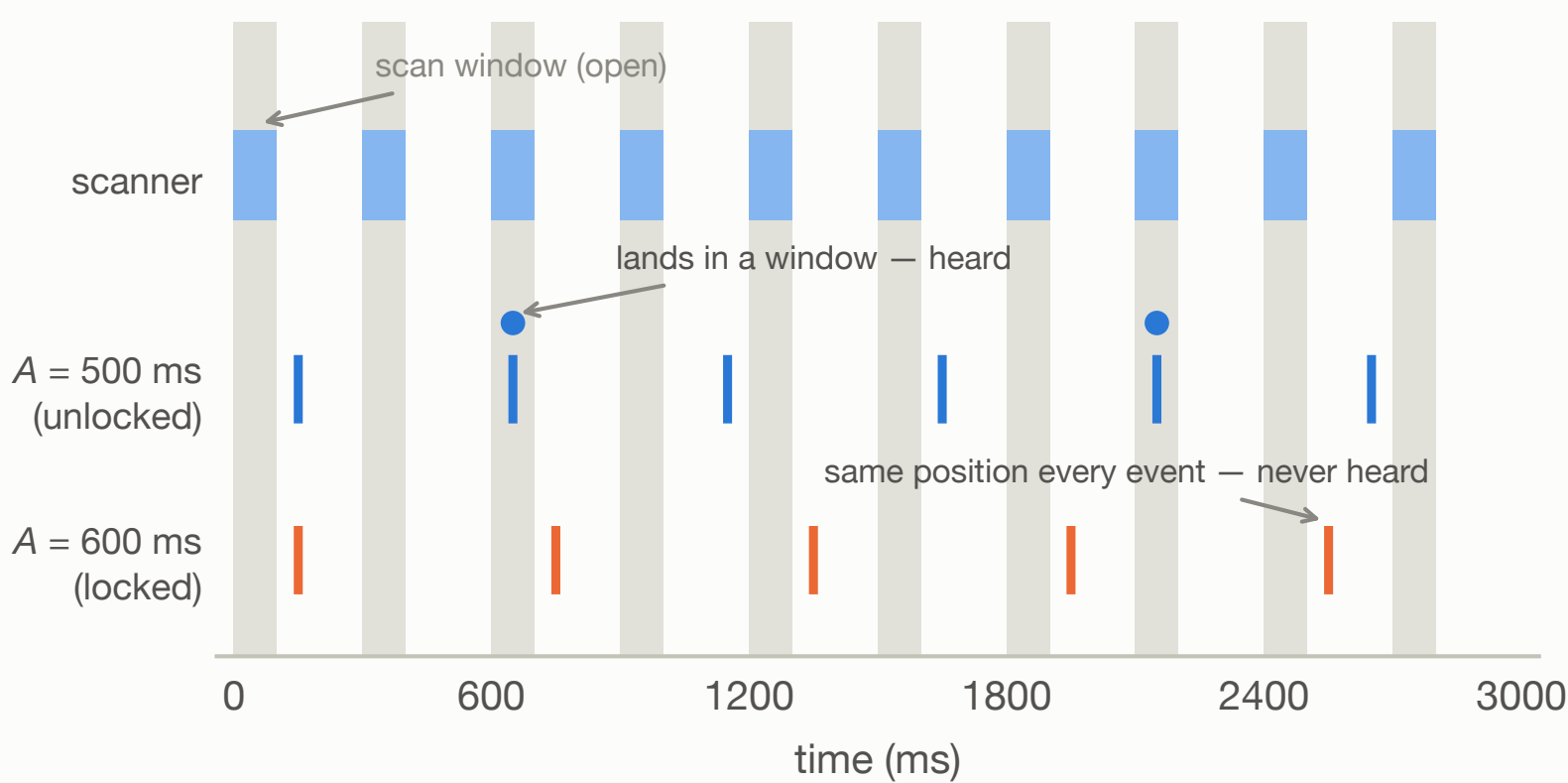


Advertising interval A , 200–1400 ms at -24 dBm.

More broadcasts cannot rescue it — at 16 per key, 98.9% at 500 ms against 76.7% at 600 ms

Why: scan-window arithmetic

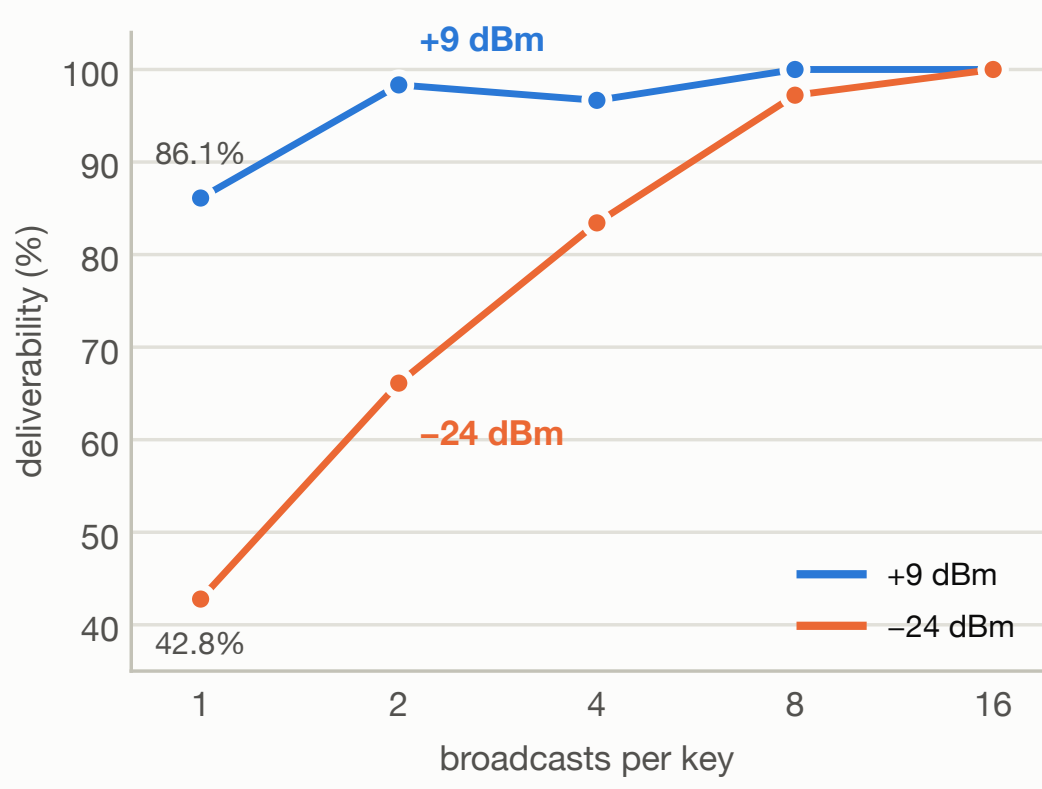
- A scanner listens once per ~ 300 ms cycle, deaf between
- A multiple of 300 ms lands in the same slot every time
- Stuck in the deaf gap, every broadcast misses alike



Scan-cycle geometry; window and offset illustrative.

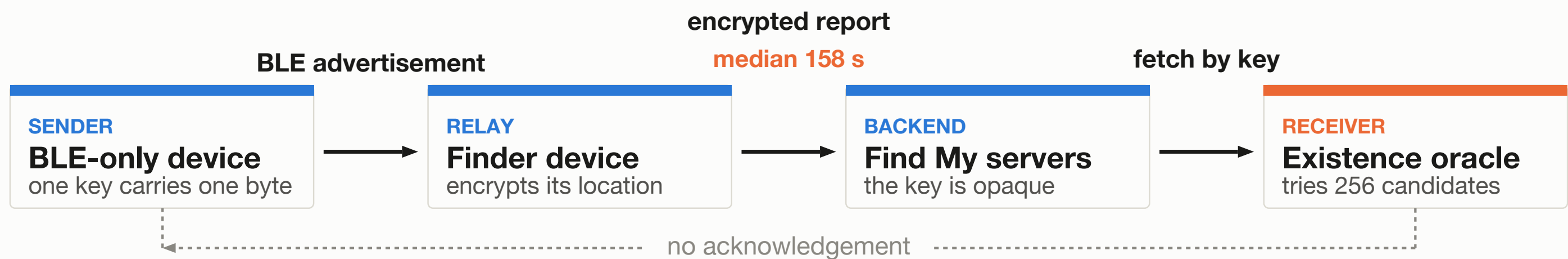
Repetition versus power

- Broadcasts per key: the strongest control
- 42.8% at one broadcast, 100% at sixteen
- Power moves the ceiling, not the shape



Broadcasts per key, at -24 dBm and $+9$ dBm.

The channel, end to end



- One byte per key: HKDF under a 32-byte uid , carrier $x(d \cdot G)$
- Receiver tries 256 candidates; the one with a report wins
- Every carrier is valid — a miss is loss, never corruption
- The uid keeps the channel private end to end

Verified on real hardware

```
Bluetooth Low Energy Link Layer
Access Address: 0x8e89bed6
> Packet Header: 0x2542 (PDU Type: ADV_NONCONN_IND, TxAdd: Random)
Advertising Address: d6:a3:14:82:eb:55 (d6:a3:14:82:eb:55)
  > Advertising Data
    > Manufacturer Specific
      Length: 30
      Type: Manufacturer Specific (0xff)
      Company ID: Apple, Inc. (0x004c)
      > Data: 1219001ff57d0a8c36a817d730c4c876a78b05e58f4d36a6d0300
      CRC: 0x9d1061
```

A sniffer parses the beacon as Apple Find My — company ID $0x004c$.

```
31 advertising bytes:
1eff4c001219001ff57d0a8c36a817d730c4c876a78b05e58f4d36a6d0300
off bytes      field
0 1e          AD length (30 = rest of this AD structure)
1 ff          AD type: manufacturer specific data
2 4c00        company ID 0x004c (Apple), little-endian
4 12          Apple message type 0x12 = Find My / OF
5 19          Find My payload length (25 bytes follow)
6 00          status byte
7 1ff57d0a8c36a817d730c4c876a78b05e58f4d36a6d0300 advertising key, bytes 6..27 (22 B)
29 03         key byte 0 >> 6 (top 2 bits, lost to 0xc0)
30 00         hint
```

All 31 advertising bytes accounted for; 22 of them carry the key.

```
> scripts/.venv/bin/python scripts/fetch_reports.py --epochs 100
Derived 100 epoch keys (0..99); p_0 = 7735d57a9fc2728202812fcd32a6b1b268773642a020b2bdc4574b3.
Restored session for
18 report(s) across 100 epoch(s) (oldest first):
[epoch 0] 2026-06-13T11:38 lat=+1.292 lon=+103.772 ±60m conf=0
[epoch 0] 2026-06-13T11:39 lat=+1.292 lon=+103.771 ±28m conf=0
[epoch 0] 2026-06-13T11:39 lat=+1.292 lon=+103.772 ±56m conf=0
[epoch 1] 2026-06-13T11:40 lat=+1.292 lon=+103.772 ±40m conf=0
[epoch 1] 2026-06-13T11:40 lat=+1.292 lon=+103.772 ±74m conf=0
[epoch 1] 2026-06-13T11:40 lat=+1.292 lon=+103.772 ±74m conf=0
```

Fetches back: 18 reports across 100 epoch keys, each a decrypted location.

What the channel is

- Erasure channel — never corrupts
- At least 8 broadcasts per key
- The nulls expose the scanners' schedule
- 99.5–99.94% delivered at full power
- Never near a multiple of 300 ms
- Delay-tolerant only — never real-time